Toolpath Generation for Face Milling of Fused Cuboids

Generated by M365 Copilot

June 7, 2026

Original Prompt

You are a software+mechanical engineer tasked with generating a toolpath generation software for solids with some fused cuboids on the same base floor but of different heights and widths and lengths. The top faces boundaries are known. How will you produce the toolpath footprint for face milling the top faces without gouging into vertical walls if they exist. The existence of vertical walls is known. The tool must enter from outside the face.

Concept Overview

Each top face is treated as a machining island at height Z_i with polygon boundary P_i. The tool of radius R must remain within a safe configuration space that avoids vertical walls.

Wall Classification

For each edge:

- If neighbor height $< Z_i$: safe drop-off
- If neighbor height $= Z_i$: merge region
- If neighbor height $> Z_i$: vertical wall (blocked)

Safe Region Construction

Define the safe machining region:

$$S_i = P_i \ominus B_{blocked}$$

where only blocked edges are offset inward by R.

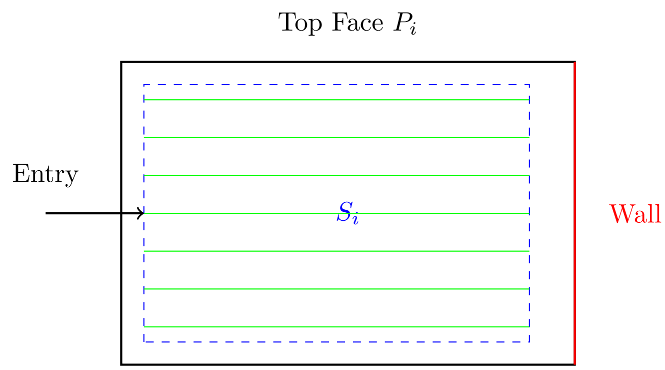
Entry Strategy

Entry edges must lie on drop-offs or outer boundaries. Tool enters via linear or arc lead-in from outside.

Toolpath Strategy

Zig-zag raster toolpaths are generated inside S_i.

TikZ Illustration



Algorithm Summary

```
for each face (sorted highest to lowest):
    classify edges
    construct safe region
    select valid entry edge
    generate raster paths
    ensure no wall collisions
```

Key Insight

This is a configuration space planning problem where walls are expanded by tool radius and paths are generated for the tool center.